2025 AMC 10A Problem 17 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 17. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 17

Problem:
Let N be the unique positive integer such that dividing 273436 by N leaves a remainder of 16 and dividing 272760 by N leaves a remainder of 15. What is the tens digit of N?

Answer Choices:

A. 0

B. 1

C. 2

D. 3

E. 4

Solution:

It is given that dividing 273436 by N leaves a remainder of 16, and dividing 272760 by N leaves a remainder of 15.

$$N \mid (273436 - 16) = 273420, \quad N \mid (272760 - 15) = 272745.$$

so $N \mid (273420 - 272745) = 675$. Thus N divides both 273420 and 675, or $N \mid \gcd(273420, 675)$.

Since $\gcd(273420, 675) = 45$, therefore, $N \mid 45$. The possible divisors of 45 are $1, 3, 5, 9, 15, 45.$ Since the remainder $16 < N$, therefore, the only possible value of $N = 45.$

The tens digit of N is

$$\boxed{\textbf{(E) } 4}.$$

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